# Goal: Test Whether Distilled CoT Improves Faithfulness

### Research question

Can one SFT pass on **DeepSeek-revised CoTs** make a 3B model's generated reasoning more aligned with the parametric knowledge supporting its answers?

### Base idea

- Student: LLaMA-3.2-3B-Instruct.
- Teacher revises student CoTs, not just final answers.
- Evaluate with **Faithfulness by Unlearning Reasoning Steps** (FUR).

### Pipeline

#### 1. Draft

Base LLaMA-3B generates OpenBookQA rationales.

#### 2. Revise

`deepseek-v4-pro` rewrites concise, answer-consistent CoTs.

#### 3. SFT + FUR

Fine-tune once, then compare Base vs. SFT on the same FUR test set.

FUR follows the paper "Measuring Chain of Thought Faithfulness by Unlearning Reasoning Steps" (arXiv:2502.14829).

# Experimental Setup

### Data and SFT

| | |
|---|---|
| Base model | Llama-3.2-3B |
| Teacher | `deepseek-v4-pro` |
| Draft set | 2000 OpenBookQA train examples |
| Accepted SFT data | 1852 examples (92.6%) |
| SFT method | LoRA, `down_proj` only |
| LoRA config | rank 32, alpha 64, dropout 0.05 |
| Training | lr $= 10^{-4}$, 1 epoch, max len 512 |

### FUR evaluation

| | |
|---|---|
| Target split | 100 OpenBookQA test questions |
| Retain split | 20 held-out questions |
| Compared arms | Base vs. DeepSeek-Rev. SFT |
| Unlearning | NPO + KL |
| FUR LR / epochs | $3 \times 10^{-5}$ / 5 |
| Updated params | `down_proj` only |
| Seed | 1001 |

**Primary readout.** Report FF-HARD/FF-SOFT on agreement questions; use the strict paired subset for uncertainty checks.

# Main Results: Accuracy, Controls, and Faithfulness

| Model | Dir. | CoT | Agree | Steps | Eff. | Spec. |
|---|---|---|---|---|---|---|
| Base | 70.00 | 73.00 | 82.00 | 6.37 | 92.58 | 94.04 |
| SFT | **73.00** | **77.00** | **85.00** | **3.12** | **97.62** | **98.79** |

| Model | Agree N | FF-HARD | FF-SOFT | Valid steps |
|---|---|---|---|---|
| Base | 82 | 70.73 | 54.09 | 495 |
| SFT | 85 | 65.88 | **56.07** | 311 |

**What changed?**

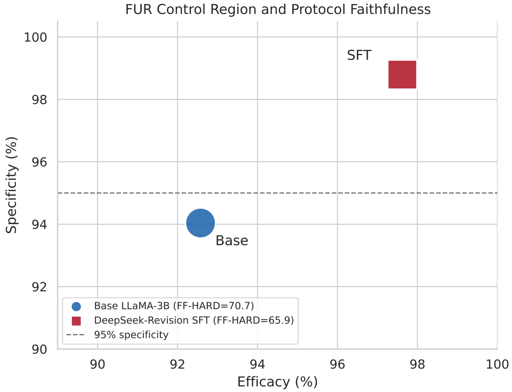
- CoTs become shorter: **6.37 to 3.12** steps.
- FUR becomes cleaner: higher efficacy and specificity.
- Faithfulness is **not clearly improved**: FF-SOFT rises, but FF-HARD falls.

**Paired check.** On 75 common agreement questions:

- FF-HARD: $-4.00$ pp, CI $[-16.00, 8.00]$.
- FF-SOFT: $+1.12$ pp, CI $[-7.95, 9.95]$.

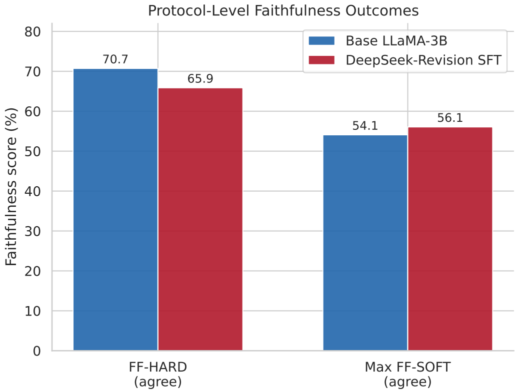
# Visual Summary

## FUR control region




SFT moves to the high-efficacy, high-specificity region.

## Protocol faithfulness




FF-SOFT improves slightly, while FF-HARD does not.

**Takeaway.** **One-shot distilled-CoT SFT improves compactness and FUR control quality.** It does **not yet establish a statistically stable faithfulness gain**.